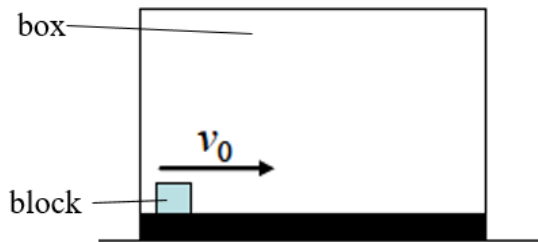


Q4



Taking right as the positive direction, the initial momentum of the box-block system relative to the floor is mv_0 . After several collisions with the walls of the box, the block comes to rest in the box. The box, however, is still moving. The final momentum of the box-block system is $(M + m)v$ where v is the final speed of the box-block system. From conservation of momentum, $mv_0 = (M + m)v$

$$\text{Or } v = \frac{m}{M+m} v_0$$

Note that although the KE of the system is not conserved, the total momentum is.

Ans: D